

Gavin McKay – Software Engineer (AI/ML Focus)

gavinmckay2022@gmail.com | (704) 258-1983 | gavinmckay.dev

EDUCATION

North Carolina State University

B.S. in Computer Science | Philosophy Minor

Raleigh, NC

Expected Graduation December 2026

Relevant Coursework: Software Fundamentals and Lab, Discrete Math, C and Software Tools, Data Structures & Algorithms, Automata & Languages, Software Engineering, Operating Systems, Intro to AI, Automated Learning and Data Analysis, Applied Web-based Client-Server Computing

EXPERIENCE & LEADERSHIP

Neurotech at NCSU | Lead Software Engineer

Sep 2025-Present

- **Engineering Leadership:** Spearheading the software development roadmap for brain-computer interface (BCI) projects, managing a team of developers to integrate electroencephalogram (EEG) data with real-time applications.
- **Machine Learning & Signal Processing:** Designing and implementing machine learning algorithms to filter raw EEG signal noise and classify specific brain-wave patterns to trigger digital and physical actions.
- **Assistive Technology Development:** Leading the creation of a grid-based, switch-accessible Augmentative and Alternative Communication (AAC) application utilizing eye-tracking technology for children with severe motor impairments.
- **Hardware Integration:** Coordinating with neurotech industry partners to integrate high-fidelity EEG headsets for future-phase projects, including brain-controlled robotics (RC cars) and neuro-feedback gaming environments.

Code Black | (Event & Company Outreach Coordinator)

Apr 2024-Dec 2024

- **Corporate Partnerships:** Directed outreach initiatives to industry representatives to secure financial sponsorships and recruitment presence for club events.
- **Event Orchestration:** Led the planning and execution of the annual Code Black Hackathon and career fair, managing logistics, vendor relations, and participant experience.
- **Strategic Collaboration:** Partnered with the Raleigh chapter of **Blacks in Technology (BIT)** to co-coordinate a collaborative professional event in observance of Black History Month.
- **Career Development:** Managed organization and tabling logistics for corporate partners during our career fairs to facilitate networking opportunities for members.

PROJECTS

Neural-Logic AAC – Real-time EEG signal processing & AI-powered assistive technology

2025-Present

- Architected an end-to-end signal processing pipeline to filter raw EEG data and map neural patterns to physical and digital actions, such as mind-controlled gaming and assistive communication.
- Engineering a communication interface designed for children with motor impairments, prioritizing low-latency signal-to-text translation.

CageIntel – AI-powered UFC analytics application

2025-Present

- Built a Python scraper using BeautifulSoup & Requests to create a comprehensive dataset of UFC fight stats, including detailed round-by-round data from ufstats.com.
- Trained a PyTorch model on the collected data to forecast winners.
- Containerizing the full-stack application to ensure consistent, isolated environments and enable CI/CD.

Compressed File Archiver (Snark) – C & Software Tools Project

2025-2025

- Developed a C-based command-line program to create and extract files from a custom-format compressed archive, similar to **tar** or **zip**.
- Implemented file compression logic to reduce archive size and managed all file I/O operations.

Password Manager – Python-based password storage application

2022-2023

- Created a Python-based password database application using MySQL and Python for the full tech stack.
- Engineered the application to store passwords locally via a hidden folder, encrypted using the SHA-256 algorithm, and accessed via a master password.
- Implemented a secure password generator for enhanced security.

SKILLS

Tools: Git/GitHub, Docker, Ollama, Azure, Oracle DB, Kubernetes, MATLAB, MySQL, TensorFlow, PyTorch, OpenCV, Tkinter, pygame, Flask, Tailwind CSS, Selenium, BeautifulSoup, Linux (Debian, Arch, Fedora/Red Hat)

Languages: Python, Java, Javascript, Typescript, HTML5, CSS3, C, C#, C++, Bash, Assembly, SQL

Frameworks: React Native, Node.js, Yarn, Django, FastAPI, RestAPI, .NET, Angular

Technical Skills: Software Development Lifecycle, OOP, Machine Learning Algorithms, Data Modeling/Visualization, Software Architecture, Testing & QA, Database Design, DevOps, Problem-Solving, Communication, CI/CD, Server Operations, Linux System Administration, Command Line Operations